**Illuminating Human Language Production via Mechanistic Interpretability of LM Circuits: Animacy Effects on Sentence Structure Choice**

Yue Li, Yan Cong, Elaine J. Francis[1]

**Introduction:** Cross-linguistic research on object relative clause (ORC) production has revealed a robust pattern: speakers prefer passives (*the baby carried by the woman*) over actives when both nouns are animate, but this preference *diminishes* with inanimate head nouns (Gennari et al., 2012; Humphreys et al., 2016). Two mechanisms have been proposed: *similarity-based competition*, where two animacy-congruent nouns interfere during encoding, and *animacy-based accessibility*, where animate head nouns independently promote passive structure. Existing behavioral data cannot adjudicate between these two mechanisms because both predict high passive rates in AA (animate head noun, animate agent noun). Recent developments with neural language models (**LMs**) exhibit *human-like* structural preferences under animacy manipulation (Hanna et al. 2023; Li et al., 2025). Crucially, LMs afford a unique opportunity: we can apply *mechanistic interpretability* methods, techniques that trace how artificial neural circuits implement higher-level computations (Futrell & Mahowald, 2025; Mueller et al., 2025,2026), to directly observe the causal architecture underlying these preferences. We take GPT-2 (Radford et al., 2019) as a computational testbed, using *mechanistic interpretability* methods (Futrell & Mahowald, 2025) to observe whether structure selection depends on each noun's animacy independently or on their joint configuration. By identifying causal architecture in a system sharing our behavioral signature, we generate testable predictions for human production, asking which theorized mechanism is computationally sufficient to produce the pattern humans show.

**Method:** We conducted three analyses on GPT-2 using 384 English sentences across four animacy configurations (AA, AI, IA, II, see Fig. 1). **Study1:** To establish that the model encodes animacy as a categorical feature (prerequisite for any animacy-based mechanism), we trained linear classifiers on 70% of nouns and tested their generalization to held-out items. **S2:** To examine how animacy shapes representations, we used embedding similarity analysis (ESA) to compare noun vectors within sentences (head vs. agent) and across structure alternations (same noun in active vs. passive). **S3:** To determine causal architecture, we used joint activation patching: simultaneously replacing both noun representations from a source animacy configuration into a target configuration and measuring changes in passive preference. Critically, we employed *bidirectional* patching (e.g., both AA→AI and AI→AA), which we used to distinguish the effect of head noun and agent noun: independent/additive contributions predict symmetric effects in both directions, while interactive/joint computation predicts asymmetry. Null controls (self-patching, within-animacy patching) validated the method.

**Results & Discussion:** Classifiers generalized to held-out nouns with 92–93% accuracy, confirming GPT-2 encodes animacy as an abstract category. ESA revealed the IA configuration produced the lowest within-sentence noun similarity (~0.68), while AA showed the highest (~0.77), showing maximal differentiation where animacy aligns with prototypical agent-patient relations. The critical test came from causal intervention. If animacy-based accessibility alone drives structure selection, head and agent animacy should contribute independently, yielding symmetric bidirectional effects. Instead, we observed *robust asymmetries*: forward patching (e.g., AA→AI) produced effects 2.4-5.4× larger than reverse directions (Fig. 2A). This revealed an AND-gate architecture where both nouns must be jointly evaluated. This result supports the view that the system computes the configuration of animacy features, *not* their independent sum. Effects were strongest in early layers (0–2), suggesting that the "animacy→structure" mapping occurs during initial encoding, consistent with psycholinguistic evidence that animacy influences early stages of grammatical encoding (e.g., in eye-tracking and EEG) (Sauppe, 2017). These findings move beyond describing model behavior to identifying a computational architecture sufficient to produce human-like patterns, suggesting that competition, not accessibility alone, underlies animacy effects in human sentence production.

1. Affiliation: Purdue University. Contact: Yue Li (li4207@purdue.edu)

**Figure 1.** Four animacy configurations with example stimuli

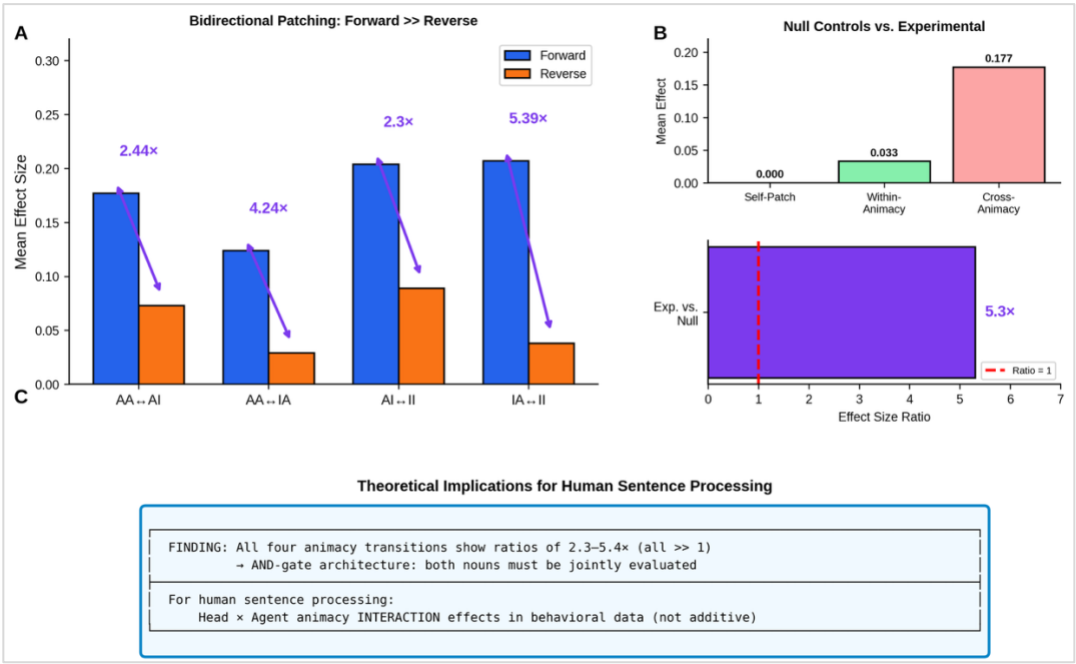


**Figure 2.** Results summary: bidirectional causal patching reveals asymmetric processing and AND-gate architecture in neural language models such as GPT-2.

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